UNIT 3 · TOPIC 9(B)

Cost per Hour, Cost per Hectare and Break-even Use

Syllabus topic: Cost analysis of animal and tractor operated implements and tractors

Worked example — tractor cost per hour

Tractor C = ₹8,00,000; S = 10% = ₹80,000; L = 10 yr; use 1000 h/yr; i = 10%.

Fixed cost item	Calculation	₹/yr
Depreciation	(800000 - 80000)/10	72,000
Interest	(880000/2) × 0.10	44,000
TIS (2%)	0.02 × 800000	16,000
Total fixed	÷ 1000 h → ₹132/h	1,32,000

Variable cost item	₹/h
Diesel: 5 L/h × ₹90	450
Lubricants: 25% of fuel	112
R&M: 8%/yr of C ÷ 1000 h	64
Wages	60
Total variable	≈ 686

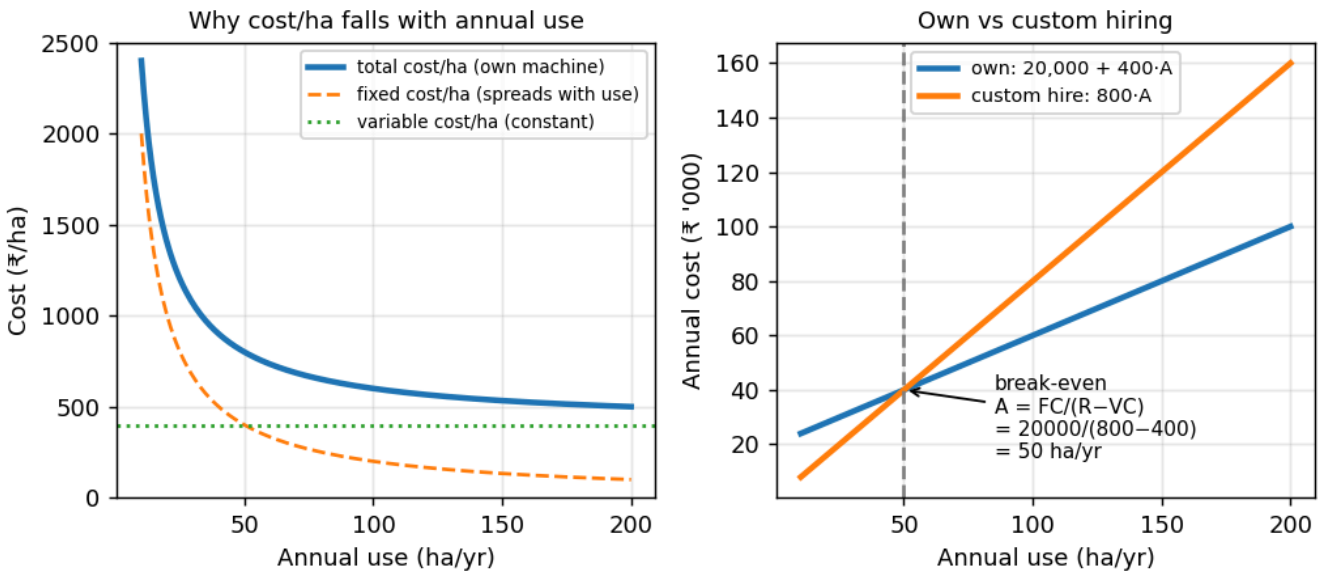
Total ≈ ₹818/h. At EFC 0.4 ha/h with a plough → **₹2,045/ha** (plus the plough's own costs).

Cost per hectare and break-even use

Cost/ha = fixed cost/ha (falls hyperbolically with annual use) + variable cost/ha (≈ constant) — so high annual use justifies ownership.

$$\text{Break-even area } A_{be} = \text{annual fixed cost} \div (\text{custom rate} - \text{own variable cost}) \text{ (both per ha)}$$

Example. FC ₹20,000/yr, custom rate ₹800/ha, own VC ₹400/ha → A_{be} = 20000/400 = **50 ha/yr**. Below 50 ha/yr, hiring is cheaper.



Why cost/ha falls with annual use, and own vs custom hiring

Practice MCQs

Q114. Tractor: C = ₹8,00,000, S = 10%, L = 10 yr. Straight-line depreciation is:

(a) ₹60,000 (b) ₹72,000
(c) ₹80,000 (d) ₹88,000/yr

Q115. Same tractor at i = 10%: annual interest =

(a) ₹40,000 (b) ₹44,000
(c) ₹80,000 (d) ₹8,800

Q116. With annual use, the FIXED cost per hectare:

(a) Rises (b) Falls
(c) Constant (d) Doubles

Q117. Break-even area for ownership vs custom hiring is:

(a) FC×(R-VC) (b) FC/(R-VC)
(c) (R-VC)/FC (d) R/VC

Q118. FC ₹20,000/yr, custom rate ₹800/ha, own variable cost ₹400/ha → break-even use:

(a) 25 (b) 50
(c) 80 (d) 100 ha/yr

Q119. Below the break-even annual use, it is cheaper to:

(a) Buy the machine (b) Hire custom service
(c) Buy two machines (d) Idle

Answers

Q114 — (b) ₹72,000 — *D = (800000-80000)/10 = 72,000.*

Q115 — (b) ₹44,000 — *Interest = (800000+80000)/2 × 0.10 = 44,000.*

Q116 — (b) Falls

Q117 — (b) FC/(R-VC)

Q118 — (b) 50 — *A_be = 20000/(800-400) = 50 ha/yr — the own-vs-hire crossover in the figure.*

Q119 — (b) Hire custom service

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 2 (Field Capacities and Cost Analysis); Hunt — Farm Power and Machinery Management; Jagdishwar Sahay; SMAM/CHC guidelines; NPTEL lectures.
